



ITSimplera Institute

Internship Report

Week 1 Task: Data Analysis & Business Understanding

Submitted by: Muskan Nazeer

Submitted to: Rayyan Khan

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Exploratory Data Analysis (EDA) on Online Retail II Dataset

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Online Retail II dataset to understand customer purchasing behavior, identify data quality issues, analyze product sales, country-wise revenue, monthly revenue trends, and generate business insights before developing a recommendation system.

Tools & Libraries Used

This project was completed using Python and various data analysis and visualization libraries. These tools were used to load, explore, analyze, and visualize the Online Retail II dataset efficiently.

Python: Programming language used for data analysis and processing.

Jupyter Notebook: Interactive development environment used to write, execute, and document the analysis.

Pandas: Used for data loading, cleaning, manipulation, and analysis.

NumPy: Used for numerical computations and mathematical operations.

Matplotlib: Used to create charts and graphs for data visualization.

Seaborn: Used to generate advanced statistical visualizations such as heatmaps and box plots.

Methodology

This project followed a systematic Exploratory Data Analysis (EDA) approach to examine the Online Retail II dataset. The analysis was performed using Python in Jupyter Notebook to understand the dataset structure, identify data quality issues, analyze product and sales performance, and extract meaningful business insights. The methodology consisted of loading the dataset, exploring its characteristics, analyzing missing values and duplicate records, calculating revenue, visualizing sales trends, and interpreting the results through statistical analysis and graphical representations.

Importing Required Libraries

The analysis began by importing the required Python libraries. Pandas was used for data manipulation, NumPy for numerical computations, Matplotlib for creating visualizations, and Seaborn for generating statistical plots. These libraries provided the necessary tools to perform exploratory data analysis efficiently.

Loading the Dataset

The Online Retail II dataset was loaded into Jupyter Notebook using the Pandas library. After successfully loading the dataset, the first five records were displayed using the `head()` function to verify that the data had been imported correctly.

```
df.head()
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

Figure 1: Preview of the Loaded Dataset

Dataset Overview

The dataset was explored by examining its dimensions, column names, data types, and summary statistics. Functions such as `shape`, `columns`, `info()`, and `describe()` were used to gain an overall understanding of the dataset before performing further analysis.

Missing Values and Duplicate Records

The dataset was analyzed to identify missing values and duplicate records. The analysis revealed missing values in the `Description` and `Customer ID` columns, along with duplicate rows. These findings indicate that data preprocessing would be necessary before applying machine learning techniques in future stages.

```
df.isnull().sum()
Invoice      0
StockCode    0
Description  2928
Quantity     0
InvoiceDate  0
Price        0
Customer ID  107927
Country      0
dtype: int64
```

Figure 2: Missing Values

```
df.duplicated().sum()
np.int64(6865)
```

Figure 3: Duplicate Rows

Revenue Calculation

A new feature named Revenue was created by multiplying the Quantity and Price columns. This calculated field was used to analyze product performance, compare country-wise sales, and evaluate overall revenue trends.

Step 7: Revenue Column

```
df["Revenue"] = df["Quantity"] * df["Price"]
```

Figure 4: Revenue Calculation

Top 10 Products by Quantity

The total quantity sold for each product was calculated and the top ten products were identified. The analysis showed that WHITE HANGING HEART T-LIGHT HOLDER was the highest-selling product based on quantity, indicating strong customer demand.

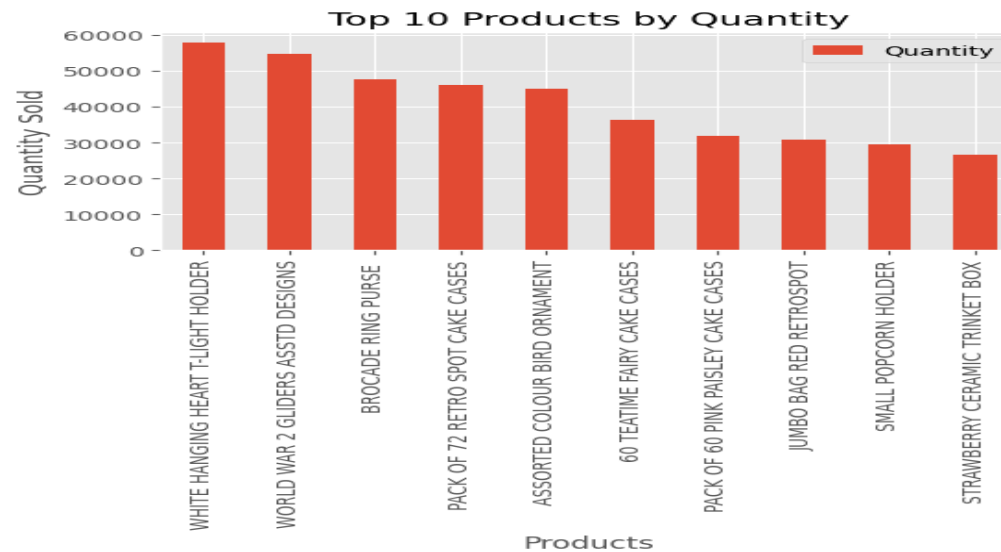


Figure 5: Top Products by Quantity

Top 10 Products by Revenue

Revenue generated by each product was calculated using the Revenue column. The analysis indicated that REGENCY CAKESTAND 3 TIER generated the highest revenue (163,051.46), followed by WHITE HANGING HEART T-LIGHT HOLDER and DOTCOM POSTAGE.

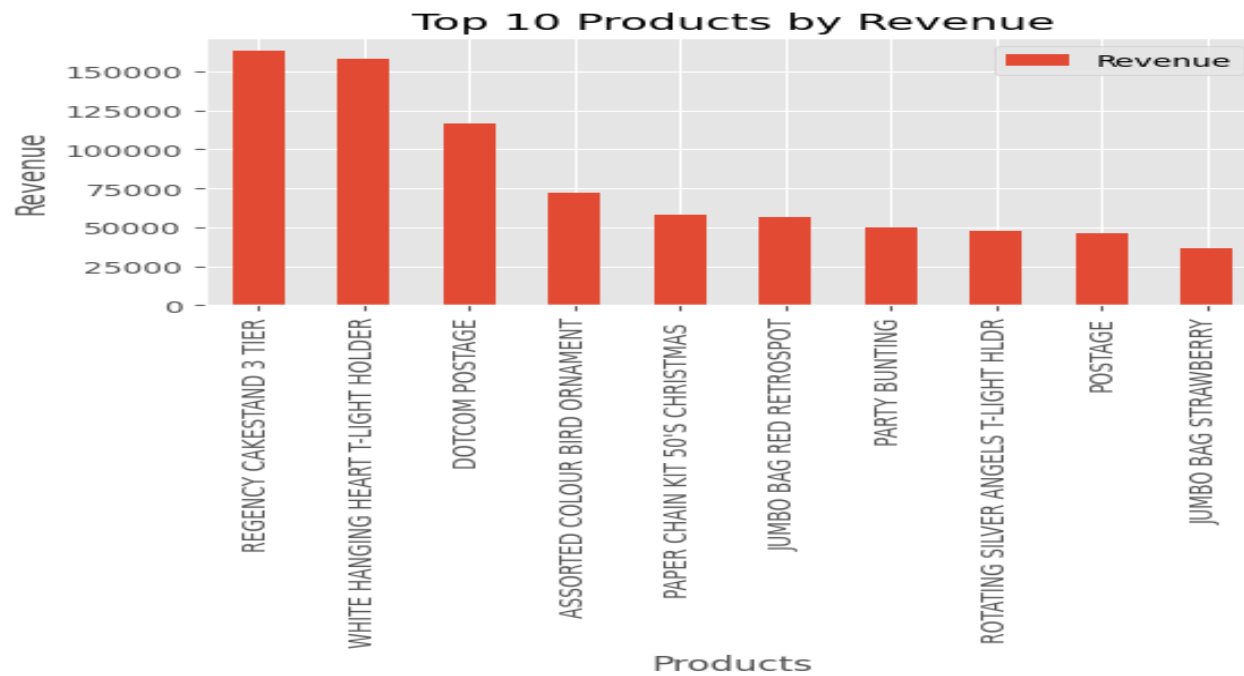


Figure 6: Top Products by Revenue

Country-wise Sales Analysis

Country-wise revenue analysis was performed to identify the highest-performing markets. The results showed that the United Kingdom generated the highest revenue, making it the company's largest customer market.

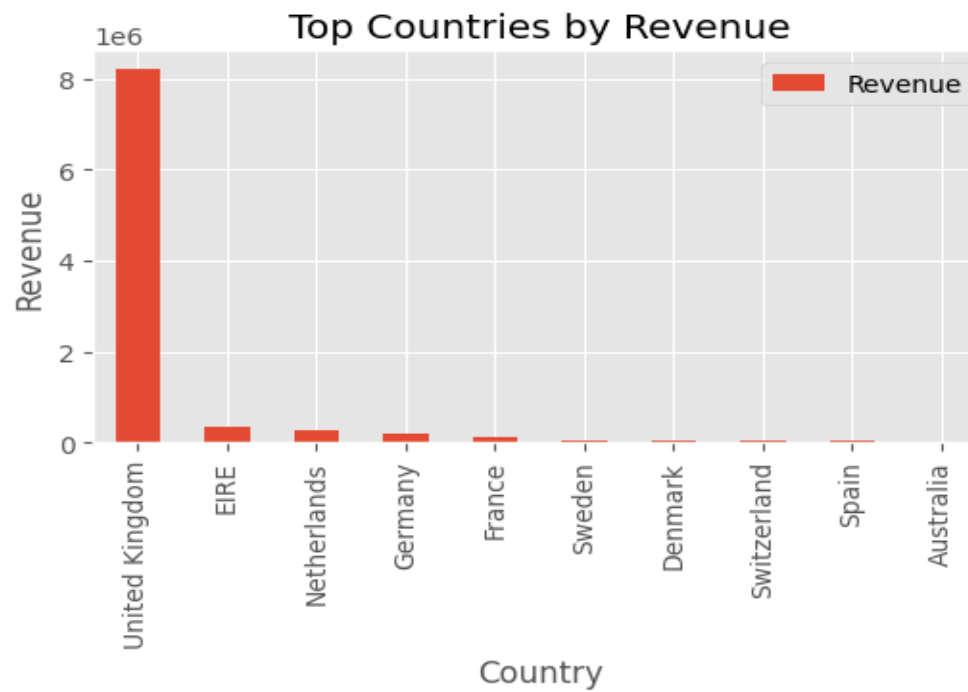


Figure 7: Top 10 Countries by Revenue

Monthly Revenue Trend

Monthly revenue was analyzed to observe sales performance over time. The line chart indicated fluctuations in monthly revenue, suggesting seasonal variations in customer purchasing behavior.

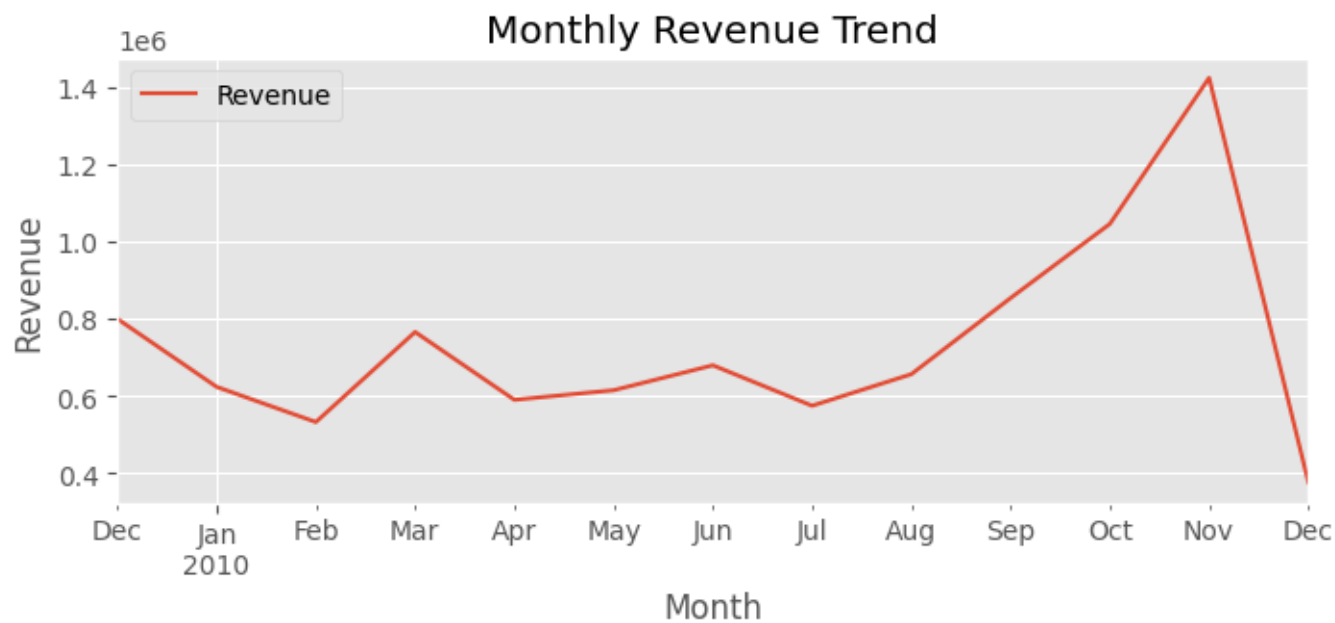


Figure 8: Monthly Revenue Trend of the Online Retail II Dataset

Correlation Heatmap

A correlation heatmap was generated to examine relationships among the numerical variables in the dataset. This visualization helped identify positive and negative correlations, providing a better understanding of the dataset's characteristics.

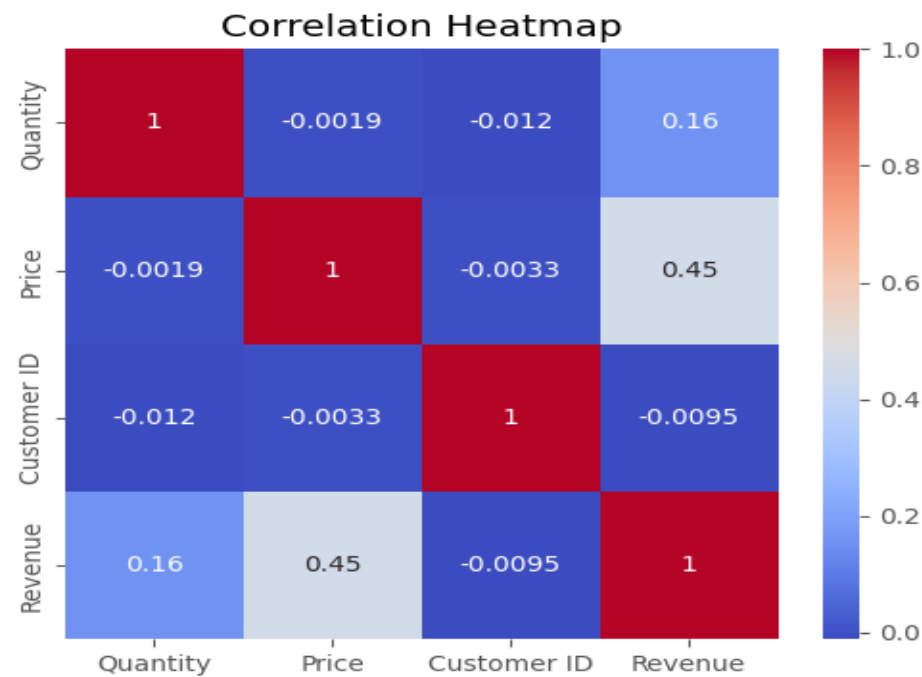


Figure 9: Correlation Heatmap of Numerical Variables

Outlier Detection

Box plots were created for Quantity, Price, and Revenue to identify potential outliers. The visualizations revealed extreme values, which are common in retail transaction datasets due to bulk purchases and high-value orders.

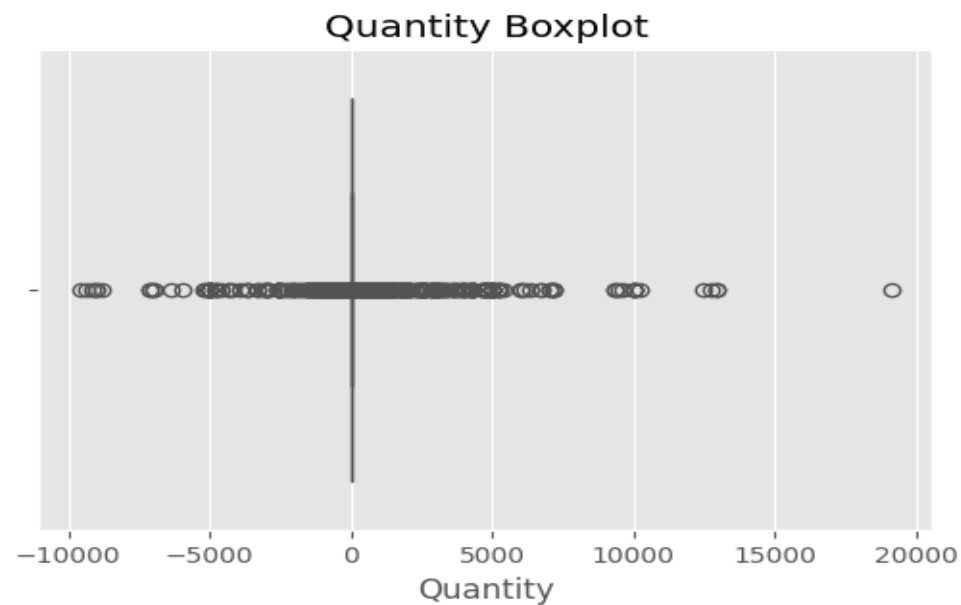


Figure 10: Box Plot of Quantity Showing Outliers

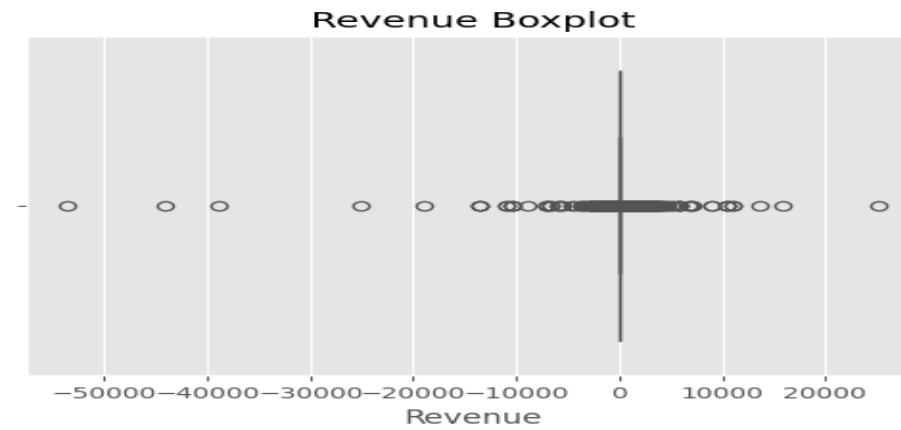


Figure 11: Box Plot of Revenue Showing Outliers

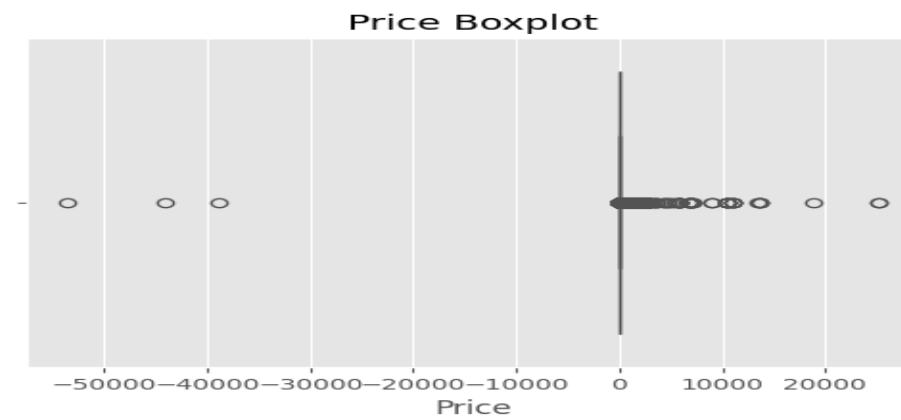


Figure 12: Box Plot of Unit Price Showing Outliers

Business Insights

The exploratory data analysis produced several important business insights:

- The United Kingdom generated the highest revenue among all countries.
- WHITE HANGING HEART T-LIGHT HOLDER was the best-selling product by quantity.
- REGENCY CAKESTAND 3 TIER generated the highest revenue (163,051.46).
- Monthly revenue showed fluctuations, indicating seasonal purchasing patterns.
- The dataset contains missing values and duplicate records that should be addressed during the data preprocessing stage before building predictive models.